Sources of Upper-Tail Human Capital: Evidence from Prussian Education Reforms

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Abstract: How can nations lead in science and technology? Education boosts human capital, crucial for economic development. This study examines whether large-scale education reforms, exemplified by the Prussia's early 19th century reforms, can cultivate the very upper-tail human capital – leading (prominent) academics. After its defeat by Napoleon, Prussia reformed its education system, emphasizing natural sciences and technology. We compare Prussian cities affected by these reforms with unaffected German-speaking cities, using a Diff-in-Diffs approach for 5-year birth cohorts over 1730–1855. Our measure of upper-tail human capital is the number of prominent scientists born in each city. Results show that after the reforms, Prussian cities produced from 36 to more than 100 scientists in 50 years. We contribute to research on human capital origins and the study of prominent individuals, offering the first systematic analysis of national education reforms in Prussia and their association with advanced skill formation. Our findings suggest positive association between scientific leadership and high-level talent.

Keywords: Prussia, upper-tail human capital, education, reform, university

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1. Introduction

How can a nation become a global leader in science and technology? Many countries seek policy recommendations on this. Investment in education can boost human capital, which is crucial for economic development and prosperity. ~~This paper examines whether large-scale education reforms, exemplified by the Prussian education reforms of the early 19th century, can foster scientific excellence by cultivating upper-tail human capital (advanced skills and knowledge).~~ **MP: This paper aims to understand whether large educational reforms can boost the development of upper-tail human capital and the scientific excellence associated with it. We use the example of the Prussian education reforms in the early 19th century.** After its defeat by Napoleon, Prussia initiated nation-wide 'Stein-Hardenberg reforms' to modernize and strengthen the state. A core component of these reforms were education reforms that focused on natural sciences and technology. The education reforms introduced a public school system, gymnasiums, and a new university model, which featured autonomy and integration of teaching and research ~~and critical thinking~~. **MP: To the best of our knowledge, this is the first paper that systematically explores the role of national education reforms in the formation of upper-tail human capital in a causal framework, with a specific focus on Prussia's experience.** We hypothesize that these reforms played a key role in developing Prussia's upper-tail human capital, bringing Prussia and Germany at the worldwide research and technological frontier within one century. **MP: I don't like this sentence, but I think this whole paragraph shouldn't be the first paragraph anyway, so I believe it will disappear.**

MP: WWe start our empirical analysis by estimating country-level production of human capital using a synthetic control design. We define Prussia as its territory within the 1820 borders and treat these borders as fixed. We approximate upper-tail human capital using the number of prominent scientists[1] as defined by Schich et al. (2014) and consider cohorts born in Prussia after 1790 to be treated. We find that, on average, Prussia was producing more scientists than the other German states. After that, we move to a difference-in-differences (DiD) design. We exploit the ~~natural~~ variation in cities' exposure to the education reforms, with scientists born in Prussian cities being affected by the reforms and scientists born in cities in other German states **MP: such as Bavaria, Saxony, Baden-Württemberg, etc. being** unaffected. We find that..... ~~For the purposes of this study, we define Prussia as its territory within the 1820 borders and treat them as fixed. We approximate upper-tail human capital by the births and deaths of prominent scientists[2] in Prussian cities and cities in other German states, including large German states such as Bavaria, Saxony, Baden-Württemberg, etc., as well as smaller states. By comparing cities in Prussia with those in the other states before and after the reforms in a Difference-in-Differences (DiD) design, we estimate the causal effects of the reforms on upper-tail human capital formation.~~

Our project contributes to two areas of economic literature: 1) the literature on the origins of (upper-tail) human capital, and its role for economic growth and 2) the studies of notable individuals, creativity clusters formation, and human capital accumulation over time and space. ~~To the best of our knowledge, we are the first to systematically explore the role of national~~

[1] Throughout the paper, we use the terms "prominent people" and "notable individuals" interchangeably.
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~~education reforms on formation of upper-tail human capital in causal framework, focusing on Prussia's experience. By studying it, we shed light on the origins of highly skilled population that contributed to Germany's scientific and industrial success in the late 19th century.~~

Previous literature has extensively documented that human capital is a strong predictor of economic growth (Mankiw, Romer, and Weil 1992; Krueger and Lindahl 2001; Barro 2001). These studies largely focus on average human capital (e.g., literacy/schooling), while the role of upper-tail human capital, i.e., the skills of the top few percent of a country's intellectual elite, has only recently gained attention. In this vein, Mokyr and Voth (2009) attribute Britain's Industrial Revolution to a minority of skilled engineers and entrepreneurs. Kelly, Mokyr, and Ó Gráda (2014) also underline the importance of highly skilled, technically capable individuals. Squicciarini and Voigtländer (2015) find that subscriber density to the Encyclopédie in 18th century France (their proxy for upper-tail human capital) predicted city growth during industrialization. Recent research on contemporaneous economies underlines the importance of math and science skills, and of entrepreneurial ability (Hanushek and Kimko 2000; Bloom and Van Reenen 2007; Hanushek and Woessmann 2012; Gennaioli et al. 2013). We contribute to this literature by documenting that education reforms can causally increase upper-tail human capital. This differs in two important aspects from the current state of knowledge. The current explanation on the origins of (upper-tail) human capital mostly attributes it to historical accident, e.g. skilled-worker immigration (Hornung 2014); or to path dependence, e.g. from historical presence of knowledge elites (Squicciarini and Voigtländer 2015) and/or from past institutions (Guiso, Sapienza, and Zingales 2016; Mokyr 2005). We are the first to emphasize that policies such as education reforms can be a cause. On the other hand, the literature studying education reforms (Duflo 2001; Banerjee et al. 2007; Chetty et al. 2011) exclusively studies the effect on average human capital, whereas we are the first to consider their effect on the upper parts of the skill distribution.

Our work also contributes to studies on the production and attraction of notable individuals and the creativity clusters formation over time and space. This literature highlights the persistence of human capital across time and geography. For instance, Borowiecki (2015) links the births of Renaissance composers in Italy to modern cultural activity, finding a strong connection. De la Croix and Licandro (2015) compile a large sample of famous people from ancient to modern times, documenting steady increases in longevity before the Industrial Revolution, suggesting that human capital contributed to the take-off of modern growth. Serafinelli and Tabellini (2022) examine the formation and decay of creativity clusters in European cities over centuries, finding that attracting creative talent is linked to institutions that promote political and economic freedoms. Schich et al. (2014) create a network of cultural centers based on the birth and death locations of notable individuals, offering a macroscopic view of human mobility and cultural attraction dynamics. Relative to the existing literature, our work provides new evidence by exploring whether prominent scientists (our proxy for advanced skills and knowledge) can be produced and attracted spatially through large-scale education reforms.

The rest of the paper is organized as follows. Section 3 describes data used in the paper; section 5 describes the main methodology employed; section 4 provides some state-level descriptive analysis of births of prominent scientists across German states; section 6 presents the

main findings, and section 7 concludes.

2. Historical background

Following Prussia's defeat by Napoleon, educational reform became part of the reconstruction of the state. Under Wilhelm von Humboldt, who headed the Section for Religion and Public Instruction in 1809–1810, reformers sought to develop general intellectual capacities before occupational specialization. The programme encompassed elementary teaching, secondary schools, and universities, although its implementation varied across institutions and localities.

The reforms produced concrete changes in teacher recruitment and school organization. The edict of 12 July 1810 introduced a state examination for prospective higher-school teachers, helping establish an academically trained profession distinct from the clergy. The Abitur regulations of 1812 specified examination requirements in languages, mathematics, and other subjects more precisely (Hinz, pp. 40–41). In Königsberg, where inadequate funding, low salaries, and inconsistent curricula had constrained higher schooling, institutional reorganization was substantially complete by late 1811. The Friedrichs-Kollegium and municipal Gymnasium acquired neohumanist orientations, while other schools were redirected toward teacher training or non-university-preparatory education (Hinz, pp. 20, 51–52). A teacher-training institution opened in 1809, and Pestalozzian methods were promoted (pp. 24–25). University reform included the foundation of Berlin University in 1810 and substantial investment in Königsberg's university (pp. 38–39).

Nevertheless, implementation departed from Humboldt's intentions. Wartime financial burdens prompted Königsberg's representatives to resist the proposed school levy; the king reduced the planned annual municipal contribution from 20,000 to 6,000 talers, and staffing and organizational arrangements were revised (Hinz, pp. 46–48). More broadly, the Gymnasium increasingly became a university-preparatory institution, while equivalent reform of town schools failed to materialize, undermining the envisioned integrated educational progression (Jeismann, pp. 395–396).

Other German states also pursued educational modernization. Bavaria introduced curricular reform in 1804 and reorganized secondary education in 1808, but financial difficulties and political opposition impeded implementation. The abolition of its real institutes in 1816 transferred responsibility for successor institutions to municipalities, making provision dependent on local resources (Buchinger, pp. 93–101). Saxony's elementary-school law of 1835 built on earlier developments and drew partly on Süvern's Prussian draft of 1819 (Moderow, pp. 86, 106, 110). Württemberg expanded state regulation while retaining confessional supervision and substantial local variation (the Württemberg study, pp. 42–47). In Hannover, higher-school supervision was reorganized in 1830, whereas rural schools remained under ecclesiastical oversight (Neugebauer, p. 43).

Prussia therefore developed an influential educational framework, but neither reform chronology nor subsequent outcomes establish uniform superiority. Clark reports primary-school attendance exceeding 80 percent by the 1840s, while identifying Saxony as achieving a comparable level (PDF p. 444). The historical comparison consequently concerns differences

in reform scope, implementation, and persistence, rather than a contrast between a reformed Prussia and unreformed neighbouring states.

3. Data

In this section, we outline the sources of data and describe how we use them to examine the role of Prussian education reforms in the formation of upper-tail human capital, with a special focus on academia, across German cities over the period 1730–1855.

3.1. Data sources

We combine three main sources: data on prominent people, data on European urban population, and data on geographical boundaries of German states in the nineteenth century.

3.1.1. Prominent people dataset

Our main data on notable individuals comes from the supplementary materials to Schich et al. (2014). The authors collected the data from the Freebase (FB) database. [3] The dataset presents "spatiotemporal birth and death information of notable individuals". It includes only deceased people, with the total dataset including 447,000 individuals.

FB includes hundreds of professional roles and specializations by genre, in addition to plain birth and death information. Schich et al. (2014) curated these roles and genres into a small set of categories, with counts of individuals in the FB database in the brackets: Performing arts (24,215), Creative (13,009), Governance (10,643), Academic (8,816), Sports (7,128), and Business/Industry/Travel (4,196). Our paper focuses on analysis of Academic category (coded in the data source as AcadEducHealth), which includes: mathematicians, physicists, scientists, chemists, historians; but also philosophers, and health and education professionals (e.g., physicians).

For robustness checks, we use the alternative panel dataset from the online appendix of De la Croix and Licandro (2015).

3.1.2. European urban population dataset

The data on population of European urban locations comes from Buringh (2021), who complements and extends the previously popular dataset on the population of European cities from 800 to 1850 in Bairoch (1988). We match birth- and death-locations of prominent people from the FB dataset to the locations in the population dataset from Buringh (2021) by geographical coordinates (latitude and longitude in degrees).

[3] Freebase.com: A community-curated database of well-known people, places, and things (Google, Mountain View, CA, 2011); www.freebase.com, as cited in Schich et al. (2014).

3.1.3. Germany state boundaries

We use shape files (coordinates of map polygons) of the German states' boundaries from the Harvard Geospatial Library, Germany State Boundaries, German Historical GIS.[4] They provide polygons for all German states for the following years: 1820, 1826, 1830, 1834, 1848, 1850, 1853, 1863, 1867, 1871, 1890, and 1914. The datasets in the shape files also contain information on state population as of the shape file year (column INFO_01).

Additionally, we use the shape file for Prussian provinces as of 1891, to conduct within-Prussia analysis of prominent people.[5]

3.2. Construction of a panel dataset

We construct a panel dataset of city-level observations for the period 1730-1855. The unit of observation is a city in a 5-year birth cohort window. We prepare the dataset in four steps.

First step. We use the Schich et al. (2014) version of the FB database on notable individuals, keeping only those who were born during the period 1701–1918.[6] Later, we shrink the sample to 1730–1855 birth cohorts for the main analysis. We also identify a list of German cities from Buringh (2021) urban population dataset (as of 1820 German boundary polygons), obtaining 297 German cities.

Second step. We match the birth locations of people born in 1701–1914 to the list of German cities from the first step, by coordinates. Initially, we performed an exact matching of birth locations and cities. However, this approach significantly decreased our sample as there were many small birth locations that were not appearing in the German cities list. Therefore, to keep all the observations, we define that a birth location matches the German city if the calculated nearest distance (using `geoinpoly` Stata command) is less than the 99th percentile of the calculated distance-to-the-nearest-neighbor distribution, similarly to approach of (Serafinelli and Tabellini 2022, p. 7).[7] By this rule, almost all birth observations and birth locations from our sample are matched to the German cities. Figure A1 in the Appendix shows the nearest distance distribution of the birth locations matched to the cities.

After cleaning of the dataset, we obtain 10,928 prominent people in all professional categories. This is our baseline sample of prominent people born in Germany.

Third step. From Buringh (2021), we attach the city-level information of the cities in our sample: level above the sea, access to the sea, and – most importntly – population (in thousands of inhabitants), which is provided with regularity of 50 years.

Fourth step. Since we have a panel dataset and the unit of observation is a city in a five-year window (birth cohort), we want to keep the observations balanced. We want to keep those cities that produce exactly zero number of prominent people (of any category) in some time windows, or in the whole period of analysis. For this, we balance our sample adding zero-number-of-prominent-people observations for such cities.

[4] https://hgl.harvard.edu/catalog/harvard-ghgis1820core

[5] https://hgl.harvard.edu/catalog/harvard-ghgis1890provinces

[6] Corresponds to the years of existence of the Kingdom of Prussia.

[7] Serafinelli and Tabellini (2022) use the 95th percentile of the distance distribution, which equals to 71km in their sample, while in our sample the 99th percentile is equal to 84,27km.

We further divide our preliminary sample into three samples subject to separate analysis:

- *Stayers*: those prominent people who were born and who died in Germany (any German state).
- *Patriots*: those whose state of birth is the same as the state of death.
- *Movers*: those whose death place is outside of Germany.

Our descriptive analysis in section 4 uses the *Stayers* sample, while our main city-level regression results in section 6 are focused on the *Patriots* sample. In the appendix, we provide regression results, figures and other analysis for the *Stayers* and *Movers* samples.

4. Descriptive analysis (state-level)

This section provides some descriptive statistics about the births and deaths of prominent people in Germany, over period 1730–1855. We compare Prussia with other largest German states, namely with the state population above 1m inhabitants as of 1820: Bavaria, Hannover, Baden, Württemberg, and Saxony.

4.1. General state-level statistics on prominent people

Figures 1 and Figure 2 provide general counts of elite (prominent) people by professional categories introduced by Schich et al. The counts are for the dataset of prominent people who were born and died within borders of Germany (as of 1820). Unfortunately, we cannot disentangle the "content" of these broad categories into narrower professional roles because we do not have the original FB dataset but rather using the dataset version from (Schich et al. 2014).

4.2. State-level descriptive statistics: Prussia and five largest states

In this section, we focus our analysis on the *stayers* sample: those prominent academics who not only were born within Germany, but also died within its borders. Although we recognize that people might have moved to other cities and states during their lifetime, the motivation behind this sample is that we want to do the least possible to . In the dataset, we cannot observe the main work country of an individual, i.e. the place where they produced the most significant achievements in their career. Therefore, to approximate this, we use what is available to us: condition on death place being in Germany, which allows us to some degree to think that these people became famous because they realized their potential in Germany.

We provide our own figure showing the dynamics of prominent individuals across the categories, born in 1730–1855 in Germany. See Appendix Figure...

Figure 3 depicts the relative contribution of Prussia to the production of prominent academics over time, compared to other states. For Prussia (the thick blue line), there can be seen a trend for a sharp rising starting around 1790 birth cohort and up until 1805 birth cohort. Prussia was not outperforming other states in term of this relative outcome. The thing is that Prussia was a large state with huge population (one order larger than population of the other largest

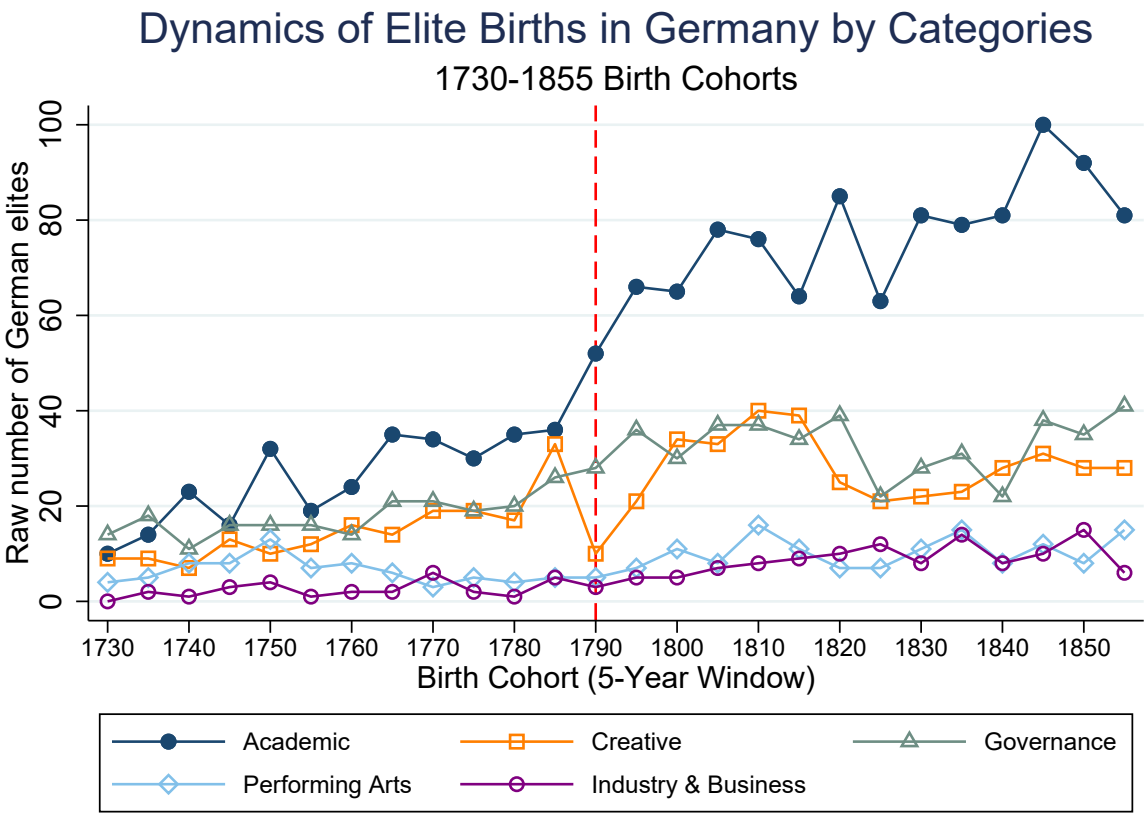


FIGURE 1

Note. Sample: prominent academics born and died in Germany.

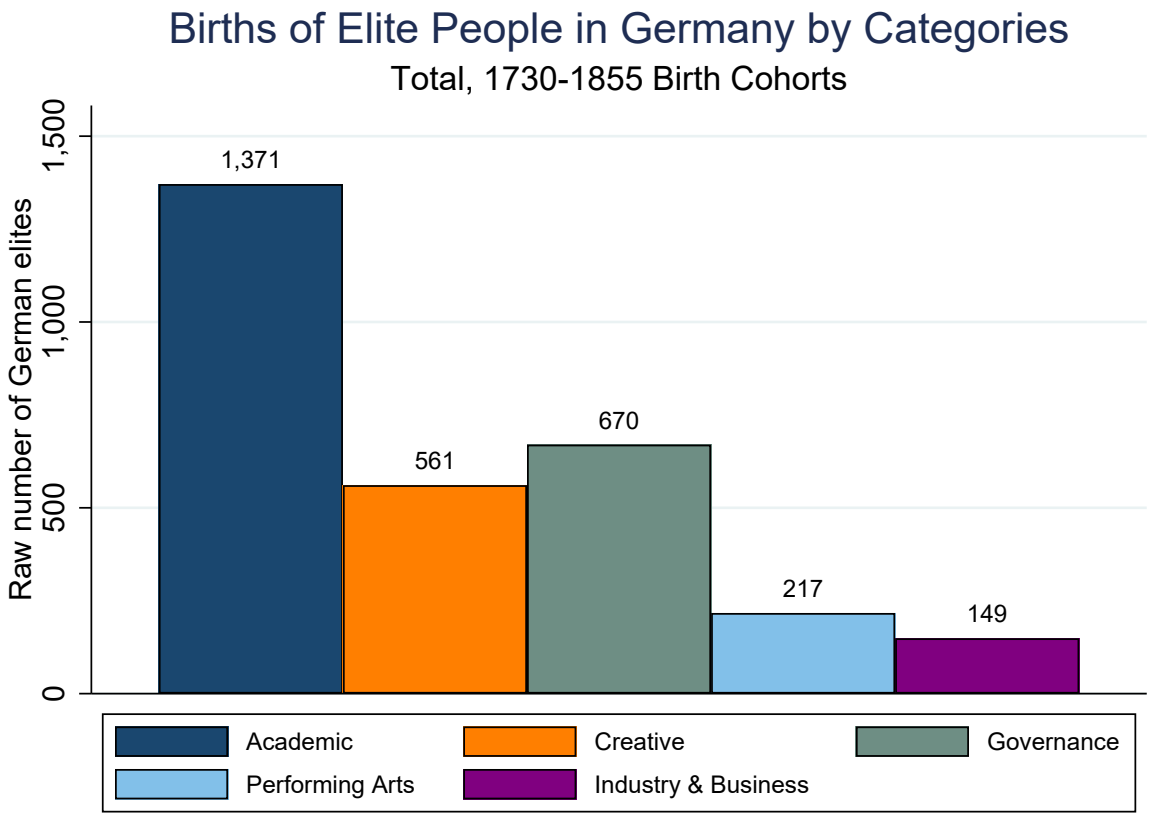


FIGURE 2

Note. Sample: prominent academics born and died in Germany.

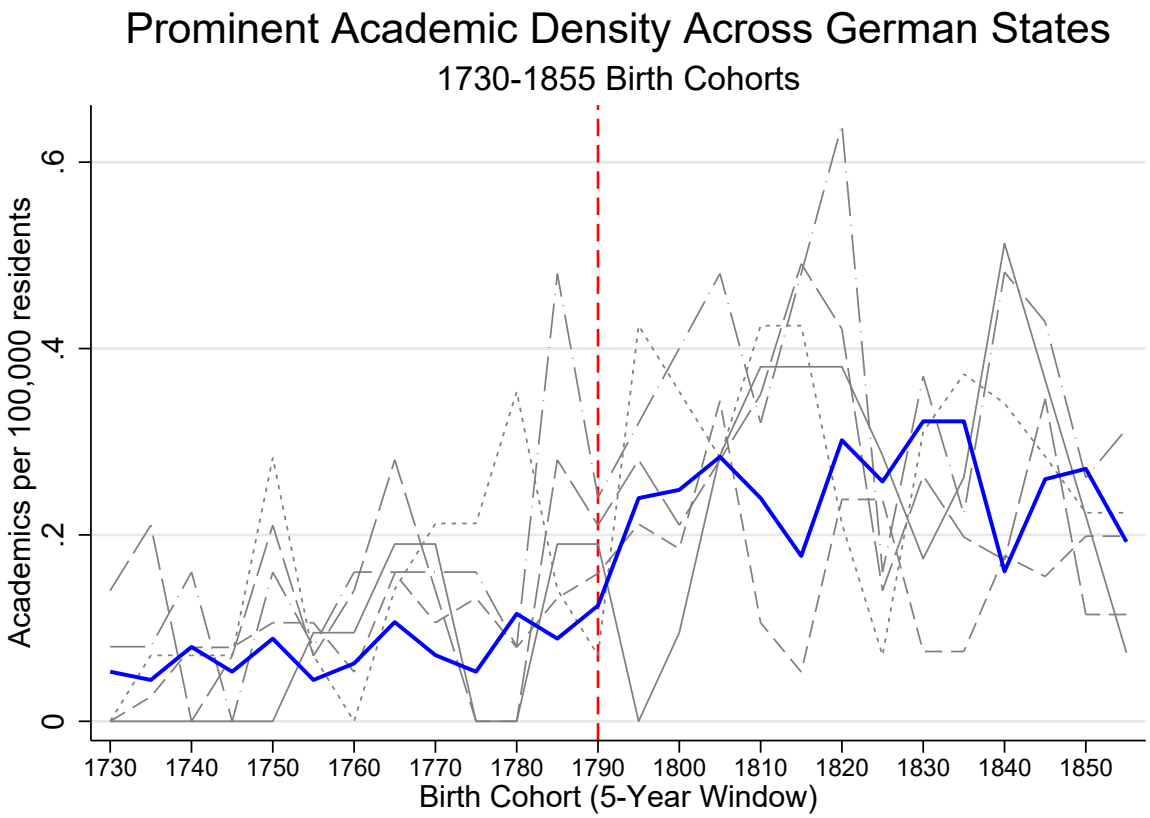


FIGURE 3. Density of elite academics across German states, 1730–1855.

Note. Sample: prominent academics born and died in Germany. We calculate the ratio of the raw number of academics to the state population, and multiply by 100,000 inhabitants. The medium thick line is Prussia, the other thinner lines are Bavaria, Hannover, Baden, Württemberg, and Saxony.

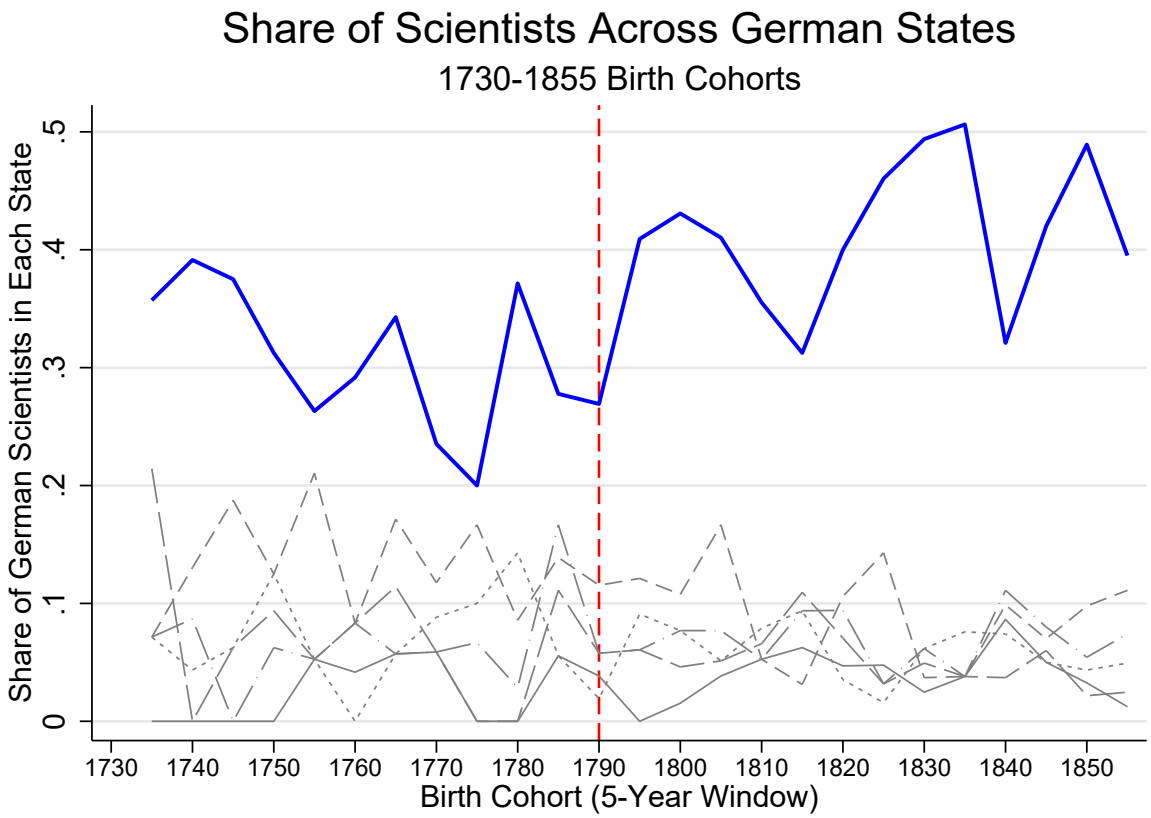


FIGURE 4. Share of prominent academics produced by states, 1730–1855.

Note. Sample: prominent academics born and died in Germany. The medium thick line is Prussia, the other thinner lines are Bavaria, Hannover, Baden, Württemberg, and Saxony. The share is calculated as the number of academics per state in a birth cohort divided by the total amount academics from all states in the birth cohort.

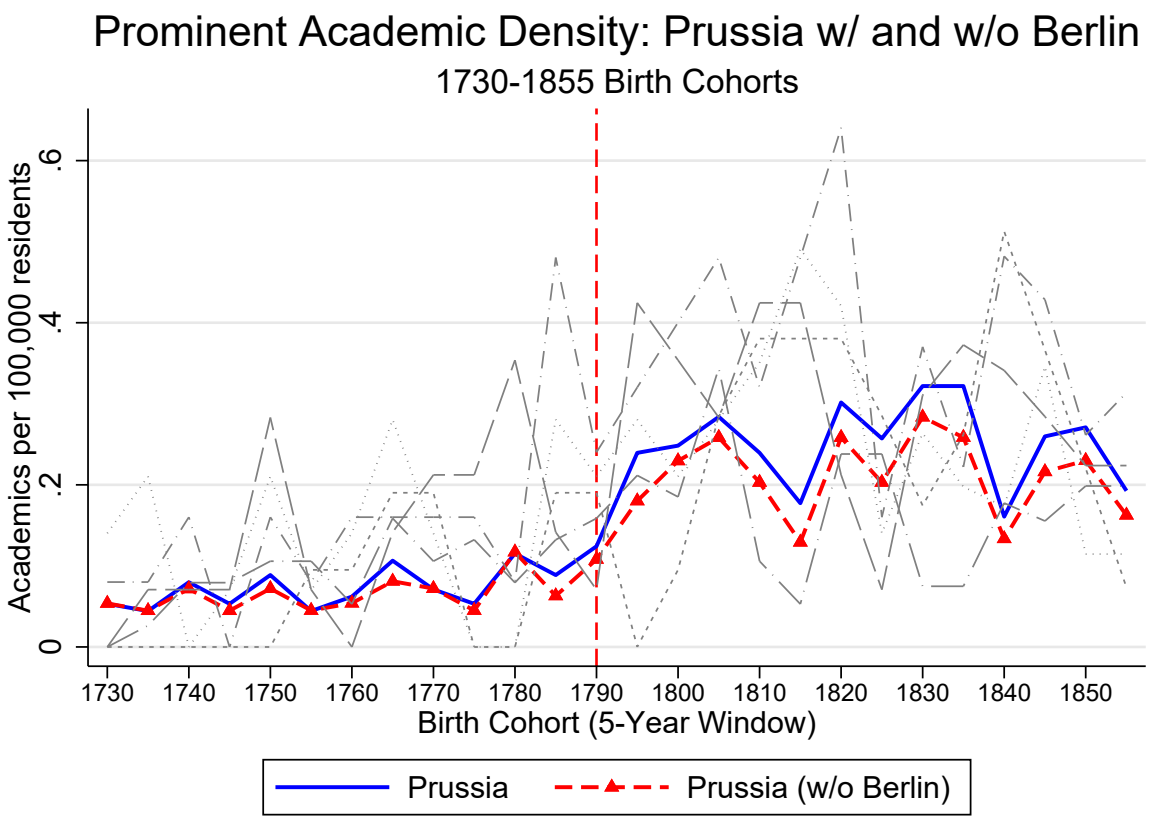


FIGURE 5. Density of elite academics across German states, 1730–1855. Prussia is w/o Berlin.

Note. Sample: prominent academics born and died in Germany. We calculate the ratio of the raw number of academics to the state population, and multiply by 100,000 inhabitants. The medium thick line is Prussia, the medium thick and dashed red line is Prussia without Berlin, and the other thinner lines are Bavaria, Hannover, Baden, Württemberg, and Saxony.

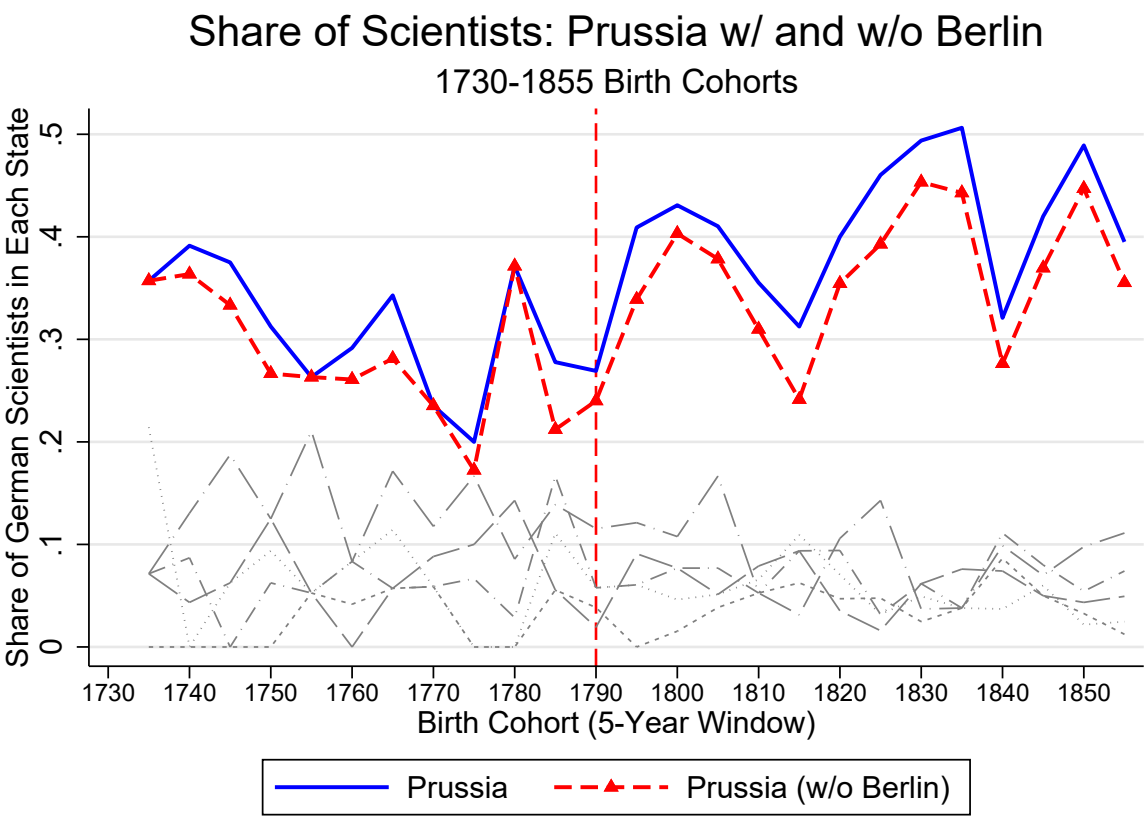


FIGURE 6. Share of prominent academics produced by states, 1730–1855. Prussia is w/o Berlin

Note. Sample: prominent academics born and died in Germany. For Prussia, data on Berlin is excluded. The medium thick blue line is Prussia, the medium thick and dashed red line is Prussia without Berlin, and the other thinner lines are Bavaria, Hannover, Baden, Württemberg, and Saxony. The share is calculated as the number of academics per state in a birth cohort divided by the total amount academics from all states in the birth cohort.

German states), hence although in absolute terms Prussia had more academics than other states, though Prussia was not the leader in the number of academics per state population. Bavaria was especially outperforming Prussia in the outcome after 1790 (dash-dot line peaks in 1820, 1840), however the observations are too noisy for the state.

Figure 4 depicts the relative contribution of Prussia in the production of total amount of academics in all largest German states, including Prussia, Bavaria, Hannover, Baden, Württemberg, and Saxony. Now we can see that Prussia actually was producing between 20% and 40% of all prominent academics in the German states before 1790. After 1790, Prussia peaked in the number of births of prominent academics up to 50% of all academics in the states around 1830. We can see that the share of elite scientists produced by other states remained quite stable even in the 19th century.

We also provide statistics without the largest city in Prussia – Berlin. Figure 5 shows the means for the scaled by state population number of elite academics for Prussia with and without Berlin, and for the other largest states. The total population of Prussia was corrected by subtracting the population of city of Berlin. It seems that excluding Berlin, the largest German city, does not eliminate the trend for a rise in births of elite academics starting from the beginning of the 19th century. Berlin seems to produce only a level effect. A similar story can be observed in the relative contribution of states: even without Berlin, Prussia still produces the largest share of scientists of all Germany, and the share is increasing starting from the 19th century, while for other states it remains relatively flat (see Figure 6).

This descriptive, state-level statistics may indicate that there was something happening in Prussia in the early 19th century what did not affect similarly the other largest German states. This allows us to hypothesize that the change in the births of prominent scientists may be related to the education reforms – and potentially more broadly to the cultural changes in Prussia – initiated by the "Stein-Herdenberg" reforms.

4.3. Synthetic control analysis: Prussia vs. five largest German states

We conduct a Synthetic Control (SC) descriptive analysis on a state-level, to demonstrate how things evolve in Prussia vs. other large German states on a macro-level. The idea of the SC is that one creates an artificial (synthetic) control group/unit as a weighted average of the outcome of untreated (comparison) states (see (Abadie, Diamond, and Hainmueller 2010) for details. This method fits our setting since we have only one treated state, Prussia, and many other untreated by the reforms states. Table 1 shows the predictor balance of the outcome between treated (Prussia) and synthetic control. Table 2 gives the weights calculated for the control states based on which the outcome for the synthetic Prussia is created. Table 2 shows that the synthetic unit is almost fully created from Bavaria (more than 90%).

In the SC analysis, our outcome of interest is the share of academics – the number of births of prominent academics scaled by the state population, per 100,000 inhabitants. The share of academics at each five-year birth cohort is our sole predictor in the SC regression. i.e. $share_academic(1730), ..., share_academic(1785)$. We also use a pre-trend variable $share_academic[_n] - share_academic[_n-1]$ in the SC regression: change in the share of academics between birth cohorts, starting from the last unaffected (according to our assumption) birth cohort, 1785. For

TABLE 1. SC: Predictor balance

	Treated unit	Synthetic control
share_academic(1730)	0.053	0.004
share_academic(1735)	0.044	0.029
share_academic(1740)	0.080	0.072
share_academic(1745)	0.053	0.074
share_academic(1750)	0.089	0.101
share_academic(1755)	0.044	0.104
share_academic(1760)	0.062	0.058
share_academic(1765)	0.106	0.164
share_academic(1770)	0.071	0.112
share_academic(1775)	0.053	0.120
share_academic(1780)	0.115	0.072
share_academic(1785)	0.089	0.140

Note. The table reports the predictor balance between the treated unit (Prussia) and its synthetic control prior to the treatment period. Predictor values are those used in the synthetic control optimization.

TABLE 2. SC: Unit weights

Control Unit	Weight
Baden	0.066
Bayern	0.909
Hannover	0.000
Sachsen	0.000
Württemberg	0.025

Note. The table reports the weights assigned to donor units in constructing the synthetic control for Prussia.

causal claims, the synthetic control unit must be matched with the treated unit as closely as possible. However, this does not to be satisfied plausibly in our data.

Figure 7 presents the graph for the Prussia and the synthetic Prussia, and Figure 8 shows the same for Prussia with data on a single largest city – Berlin – excluded. Our logic is that the large city such as Berlin could be driving the increase in academics. However, after excluding Berlin, results do not change much which supposes that contribution to elite academic births is not coming only from Berlin.

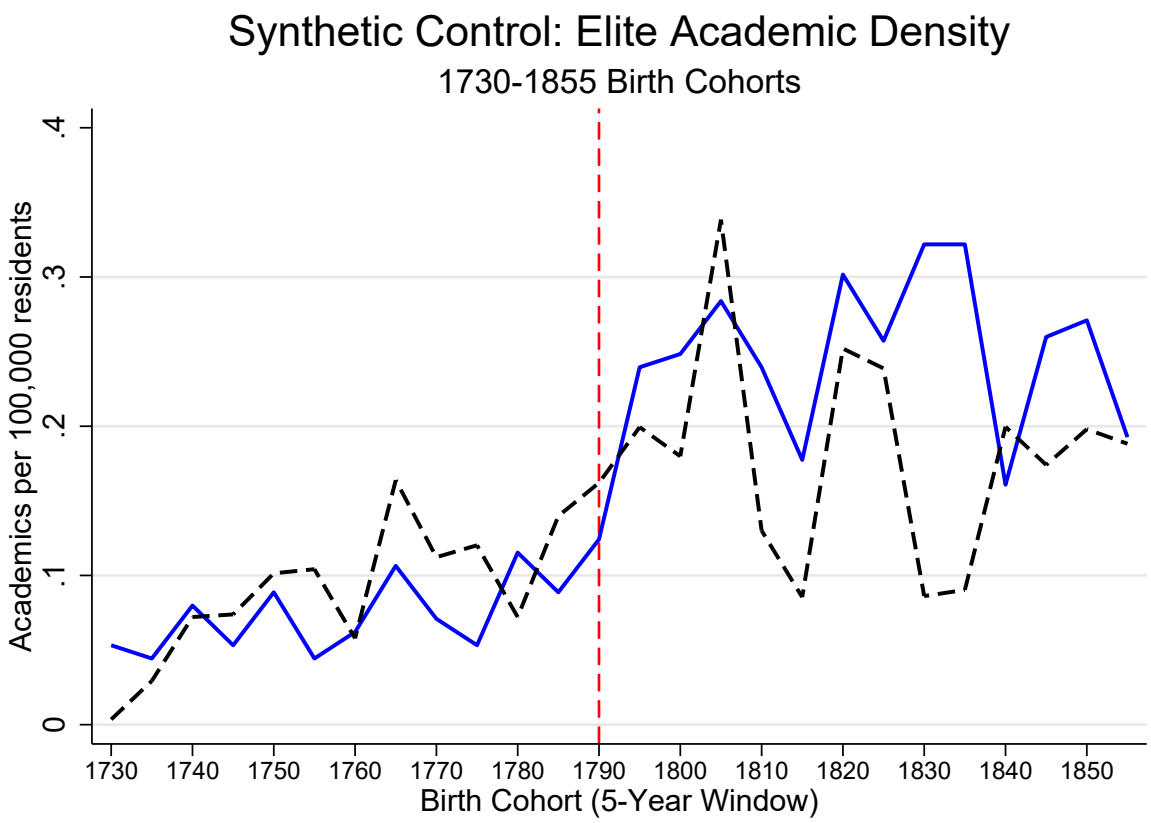


FIGURE 7. Synthetic Control: Elite Academic Density, 1730–1855

Note. Sample: prominent academics born and died in Germany. Prominent Academic Births: Prussia and Synthetic Control. Prussia is the med-thick blue line. Synthetic Prussia is (with weights in brackets): Bavaria (0.909), Baden (0.066), Württemberg (0.025), Hannover (0.000), Saxony (0.000). Predictors: share of academics per 100,000 residents by each five-year birth cohort. Num of obs in the SC regression: 156. Unit of obs: state in 5-year birth cohort window.

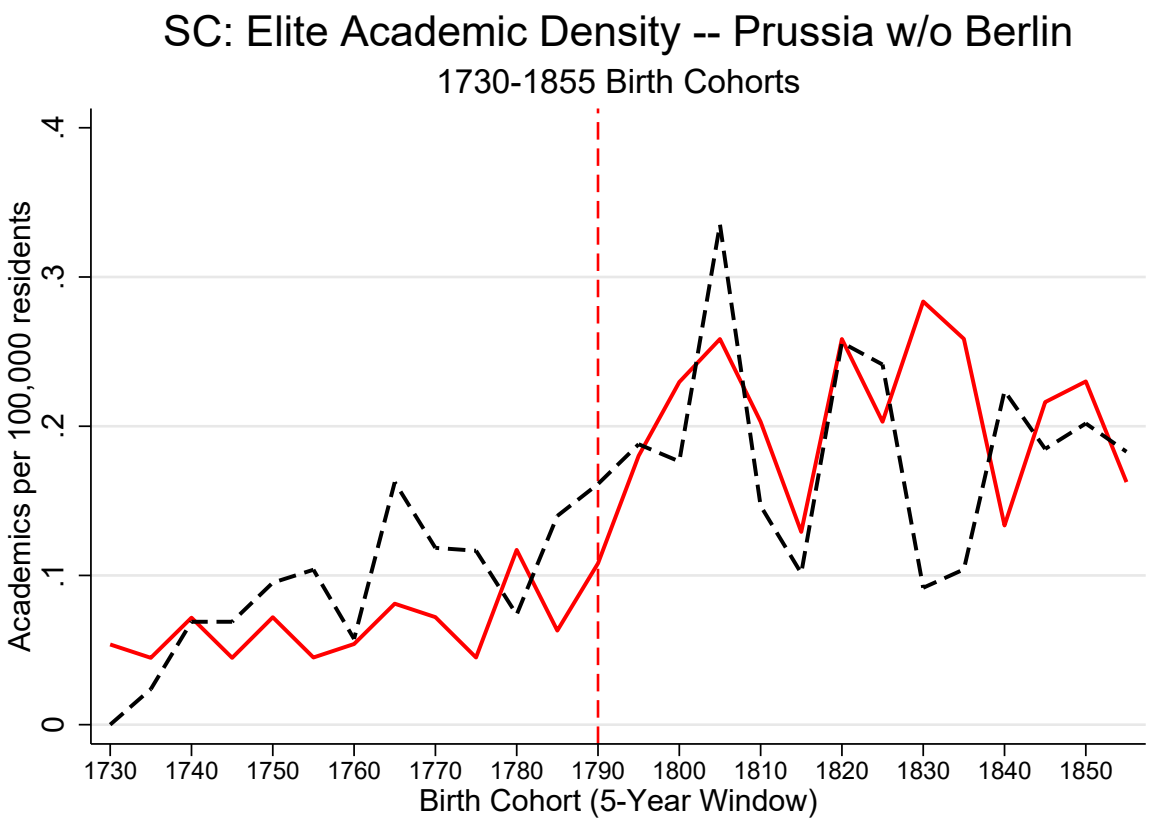


FIGURE 8. Synthetic Control: Elite Academic Density w/o Berlin, 1730–1855

Note. Sample: prominent academics born and died in Germany. Prominent Academic Births: Prussia and Synthetic Control. Prussia is the med-thick blue line. Synthetic Prussia is (with weights in brackets): Bavaria (0.909), Baden (0.066), Württemberg (0.025), Hannover (0.000), Saxony (0.000). Predictors: share of academics per 100,000 residents by each five-year birth cohort. Num of obs in the SC regression: 156. Unit of obs: state in 5-year birth cohort window.

4.4. State-level descriptive stats: other professional categories

In the Appendix, we provide state-level descriptive stats on the other professional categories of prominent people in Germany: Creatives, Governance, Performing Arts and Industry and Business. For details, see Appendix **??**

4.5. City-level descriptive statistics

TABLE 3. Correlation between city population and elite academic births

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth)			
Sample: All prominent academics born in German states 1730-1855			
NLog(Pop)	0.205***	0.204***	0.159***
	(0.033)	(0.036)	(0.046)
Constant	-1.609***	-1.597***	-1.206***
	(0.289)	(0.309)	(0.403)
City-FE			✓
Time-FE		✓	✓
# Obs	5,288	5,288	5,288
R^2	0.037	0.053	0.197

Note. Column 1 is unconditional correlation. Column 2 is conditional on time fixed effects (5-year interval). Column 3 is within-city relationship. Regressions run only on cities with 5p <= city population <= 95p (percentiles calculated on all panel).

The Figure 9 exhibits weak but clear positive conditional relationship between the city population and the births of prominent academics born in the German cities. Although the relationship is noisy, larger within-city population increases are systematically associated with higher numbers of elite academic births, consistent with the regression estimates. A 10% increase in city population is associated with approximately 0.01 additional elite academic births ($0.102 \times \text{nlog}(1.10) \approx 0.0097$), holding constant city and cohort fixed effects.

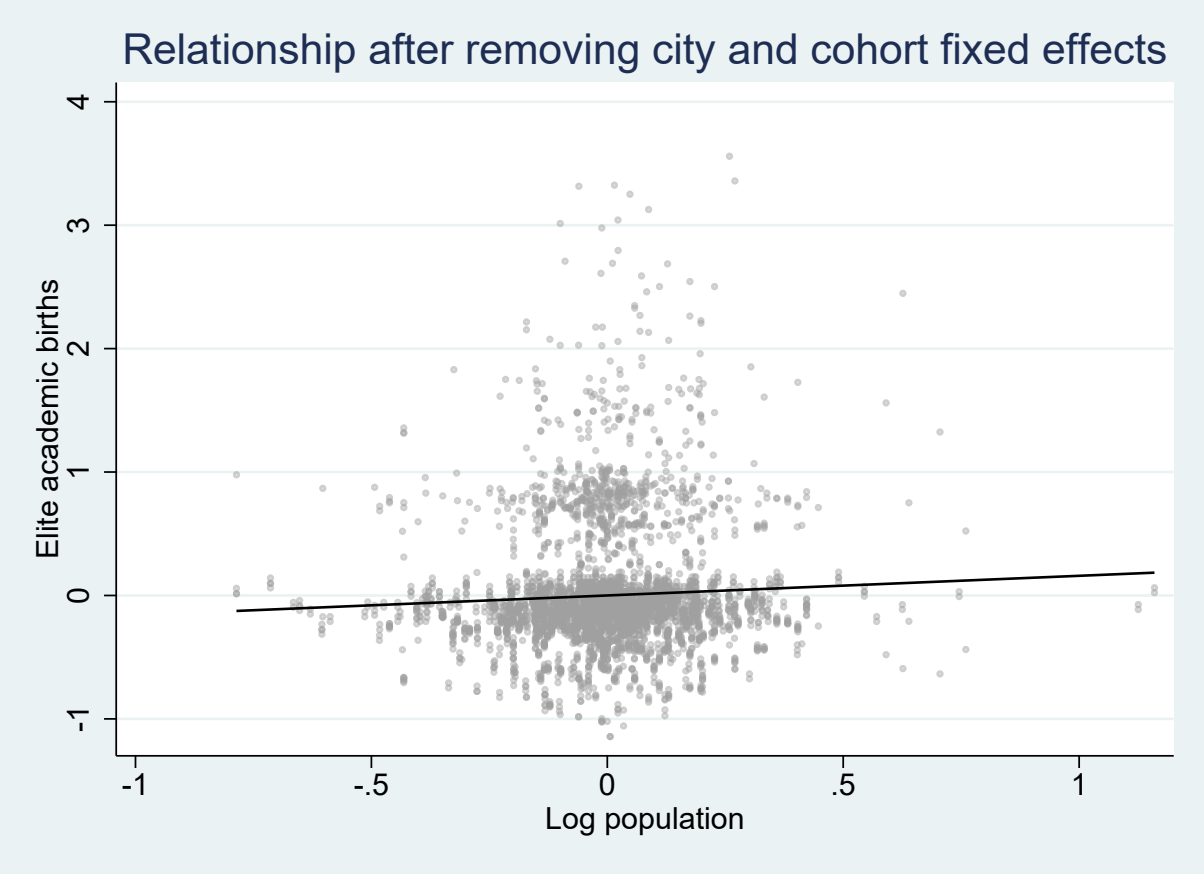


FIGURE 9

Note. Conditional correlation between German city population and prominent academic births, 1730-1855 birth cohorts.

5. Estimation Methods

In this section, we describe empirical methods to identify and estimate the effect of Prussian Reform Movement on the formation of upper-tail human capital, as proxied by the notable individuals in academic and other professional categories.

This research quantitatively identifies the effects of early 19th-century Prussian education reforms on the formation of upper-tail human capital in Prussia. For this, we compare Prussian cities (as the treated group) with cities in other German states (as the control group) by employing a Difference-in-Difference for a city-level analysis. For the purposes of this study, we define Prussia as its territory within the 1820 borders[8], treating them as fixed. The key variable of interest is the number of prominent scientists who were born in Prussia after the education reforms. We anticipate that, following the reforms, significantly more prominent scientists were born in Prussian cities compared to cities in the control states. Below, we provide more detail on each of the following aspects: data underlying our analysis, and its sources; and empirical strategy.

We consider the Prussian education reforms of early 19th century as the treatment in this study. These reforms applied across Prussia, affecting all cities within its territory. For German cities outside of Prussia, the treatment variable is set to 0. For the control group, we consider cities in other German states, including Bavaria, Saxony, Baden-Württemberg, Austria etc.

Our primary dependent variable is the number of births of prominent scientists. We define "prominent scientists" as notable individuals who worked in scientific professions (mathematics, chemistry, physics, medicine, or education). Our primary dataset for identifying these individuals comes from the work of (Schich et al. 2014), which includes detailed information on birth- and deathplaces, as well as occupations. Additionally, we plan to incorporate data from Serafinelli and Tabellini (2022), who created a dataset of famous Europeans. The number of births serves as an indicator of a city's ability to "create" prominent scientists. This data is collected at the city level, for regions including Prussia and other states. The unit of observation in our analysis is the number of prominent scientists born in a city in a 5-year time window.

Confounding factors may contribute to the formation of (upper-tail) human capital in the cities. We use data on city population from (**?**), which is our key covariate in regressions.

We analyze the impact of the Prussian nation-wide education reforms using the Difference-in-Differences (DiD) framework. Prussian cities were affected by the education reforms, while cities in other German states were not, creating a natural experiment to study the influence of the reforms on upper-tail human capital, measured by the number of prominent scientists associated with each city.

We analyze the number of births of prominent scientists in Prussian and other German cities using the DiD approach (for details, see Roth et al. (2023)). It accounts for time and city-level effects, thus helping to account for much of the variance in the outcome. The key identifying assumption of the approach is the parallel trends assumption, which we test using an event study design (the absence of treatment effects in the pre-treatment period, i.e., before the start of the reforms). Our baseline econometric model is as follows:

[8] To locate the cities belonging to states, we use a polygon shape file (.shp) of the territory of Prussia and other German states as of 1820.

$$Y_{it} = \beta \cdot \text{Prussia}_i \times \text{Post-1790}_t + \gamma \cdot \text{NLog(Pop)}_{ic_t} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} is the outcome of interest, namely the number of births of prominent scientists as defined above; Prussia_i is an indicator for a Prussian city; Post-1790_t is an indicator for the post-reform period (equal to 1 for birth cohorts in or after 1790 and 0 otherwise); NLog(Pop)_{ic_t} is the urban population of city i at the beginning of century c corresponding to birth cohort t; and α_i and δ_t denote city and time fixed effects, respectively. We estimate the model using the `reghdfe` command in Stata, which allows us to absorb multiple levels of fixed effects (city, time, and state).

6. Results

This section provides main empirical findings from the estimation of the effects of Prussian education reforms on the on the production of (upper-tail) human capital approximated by the number of births of prominent academics (scientists). We measure the number of births by calculating the city-level births of prominent people for which there is a value of 1 in the category `AcademHealthEdu` in the Schich et al. (2014) database. The key professional roles and specializations included in the category are mathematician, physician, philosopher, scientist, physicist, chemist, and teacher and education/health roles. Due to a high dispersion of the number of births of prominent scientists by years, we aggregate the number of births by a 5-year birth cohort. Our baseline regression is a Diff-in-Diff regression, which includes fixed effects on city and time. We compare change in the births for Prussian cities vs. relative change for non-Prussian German cities over time. Our main estimation sample is Patriots: those prominent scientists whose state of death is the same as their state of birth in Germany.

Regression estimates. Table 4 presents the estimated coefficients from the Diff-in-Diff regressions on the main sample, which we call Patriots: academics whose birth state is the same as their death state. All regressions include city and 5-year fixed effects.

We find that after 1790, Prussian cities experience a positive and significant effect on the number of births of prominent academics, compared to control cities in other German states, where large-scale education reforms did not occur at the time. In the Panel A (Baseline), the coefficient is about 0.03. To translate the effect, this means 0.3 prominent academics born in ten five-year birth cohorts, i.e. over 50 years. Since there are around 120 Prussian cities in the sample, this means that $0.3 \times 120 = 36$ prominent academics were born over the 50 years of all Prussia following the reform.

The effect is even larger in magnitude for the other panels, Panel B where we condition on city ever producing an academic over the period of analysis, and Panel C where we study only cities with population as of 1790 above 5,000 inhabitants. In the panel B, the effect is around two times larger (plus more than 70 academics per 50 years in Prussia), and in the Panel C – more than three times larger than in the baseline panel (plus more than 100 academics per 50 years in Prussia).

Parallel trends. Figure 10 presents a parallel trends graphical diagnostic. A key identifying assumption is that the treated (Prussian) and control (other German) cities have parallel trend dynamics in the number of births of prominent academics before the start of the reforms.

TABLE 4. Effect of Prussian education reforms on academic births (1730–1855)

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth) *Patriots sample: Academics whose birth state is the same as their death state.*			
Panel A – Baseline			
Prussia × Post-1790	0.036***	0.031**	0.031**
	(0.012)	(0.012)	(0.013)
NLog(Pop)		0.054***	0.054***
		(0.011)	(0.015)
Constant	0.013	-0.435***	-0.435***
	(0.009)	(0.084)	(0.127)
City-FE	✓	✓	✓
Time-FE	✓	✓	✓
# Obs	7,722	7,650	7,650
R^2	0.290	0.291	0.291
Panel B – Cities produced an elite academic ever			
Prussia × Post-1790	0.068***	0.057***	
	(0.016)	(0.015)	
NLog(Pop)		0.079***	
		(0.011)	
Constant	0.113***	-0.577***	
	(0.004)	(0.098)	
City-FE	✓	✓	
Time-FE	✓	✓	
# Obs	4,940	4,900	
R^2	0.276	0.278	
Panel C – Cities with baseline pop (1790) above 5,000 residents			
Prussia × Post-1790	0.113***	0.101***	0.101***
	(0.022)	(0.022)	(0.033)
NLog(Pop)		0.089**	0.089***
		(0.032)	(0.033)
Constant	0.030	-0.770**	-0.770**
	(0.018)	(0.277)	(0.310)
City-FE	✓	✓	✓
Time-FE	✓	✓	✓
# Obs	2,626	2,626	2,626
R^2	0.352	0.353	0.353

Note. DiD regression absorbing city and birth cohort fixed effects. Outcome: raw count of births of prominents in category AcadEducHealth. S.E. are clustered by state (27 clusters). Column 3 is a wild bootstrap regression (200 reps). Prussia is an indicator for a city in the Prussian border.

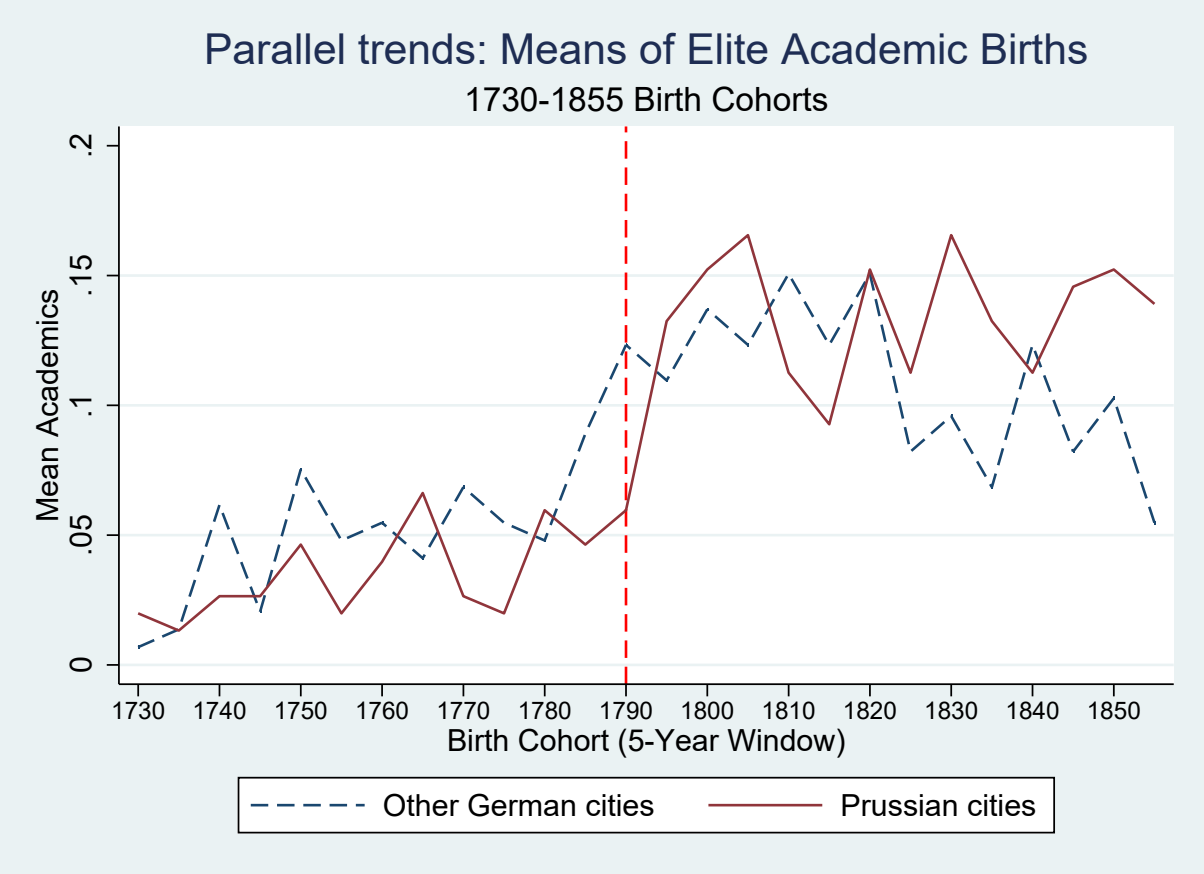


FIGURE 10. Parallel trends in elite academic births, 1730–1855.

Note. Patriots sample: academics whose birth state is the same as their death state.

Event study.

6.1. Robustness

Their dataset is based on the data from the *Index Bio-bibliographicus Notorum Hominum (IBN)*. The dataset contains dummy columns that identify 77 European cities; columns for nationalities (e.g., german) and two sets of professional occupations: one is macro-level, broad occupation (e.g., scientist), and another one is more detailed. Their observations – deceased prominent people – are bunched into birth decades. This information is enough for us to conduct a robustness checks analysing the births of prominent scientists in the few largest Prussian vs. few largest other German cities, over the birth decades 1730-1855. For details, see the Appendix E.

7. Conclusion

In this paper, we analyze the impact of the Prussian nationwide education reforms using the DiD framework. We find that the education reforms have a positive and significant effect on the production and attraction of upper-tail human capital (as measured by the number of births and deaths of prominent scientists, respectively) in Prussian cities, compared to synthetic

(comparison) cities in other German states where large-scale education reforms did not occur. Results show that after the reforms, Prussian cities produced about 250 notable scientists in 50 years, versus 150 in the prior 100 years. To the best of our knowledge, we are the first to systematically explore the role of national education reforms in the formation of upper-tail human capital within a causal framework, focusing on Prussia's experience. By studying this case, we shed light on the origins of the highly skilled population that contributed to Germany's scientific and industrial success in the late 19th century.

Discussion. Our results face several limitations which we briefly outline in this paragraph.

We identify five limitations of our results and outline how we plan to address them in future work. First, a question may arise regarding the use of traditional DiD methods. The traditional DiD methods may not yield precise estimates if there is substantial heterogeneity (e.g., in terms of population structure, religiosity, etc.) among cities in other German states compared to the relatively more homogeneous Prussian cities. We believe that a synthetic control group of other German cities would be more comparable to Prussian cities. Therefore, the use of the Synthetic Difference-in-Differences (Synthetic DiD) approach can be more advantageous. In the next steps, we plan to analyze the number of births of prominent scientists in Prussian and other German cities using the Synthetic Difference-in-Differences approach proposed by Arkhangelsky et al. (2021).

This approach merges the traditional DiD framework with the Synthetic Control (SC) approach proposed by Abadie, Diamond, and Hainmueller (2010). Synthetic DiD creates a synthetic counterpart for Prussian cities. This is done by reweighting the unaffected German cities to make their time trend parallel to the group of affected Prussian cities before the reforms. Then a traditional DiD analysis is applied. Like SC, this approach reweights and matches pre-reform trends to weaken the reliance on parallel trend type assumptions. Like traditional DiD, the Synthetic DiD accounts for time and unit level effects, thus explaining much of the variance in the outcome.

Second, Prussia may differ substantially from other German states in few characteristics, e.g., geography. This is because some German states (e.g., Bavaria) covered vast areas with varying conditions. This could create an imbalance between our treated units (Prussian cities) and control units (other German cities). To mitigate this, we will apply the Synthetic DiD approach to cities near the Prussian border, ensuring greater similarity between treated and untreated cities in terms of relevant (geographical) characteristics. Alternatively, we may apply a Regression Discontinuity design to the cities near the Prussian border, using distance to the Prussian border as a geographical instrument to predict whether a city was affected by the education reforms (for details, see Dell (2010)). These may decrease the external validity of our results, but the estimates will likely be more precise.

Third, the control states may have implemented similar education reforms during the study period. If so, this could influence our data and results, reducing the difference in the production of prominent scientists between Prussia and these states. In such a case, we would expect the gap in outcomes to narrow over time. To address this, we plan to analyze the numbers of births of prominent scientists in each state individually over the years, allowing us to account for any

similar reforms and their potential effects on our results.

Fourth, there might be confounding effects of other reforms conducted in Prussia at the same time (e.g., military reform, administration reform), which could bias our estimates of the education reforms' effect upward. We hypothesize that these other reforms may have contributed to human capital production, but it is less certain whether they directly influenced the upper tail of human capital (people with advanced skills and knowledge, e.g. scientists or engineers). To make progress in addressing this concern, we plan to analyze the births of prominent individuals across different professional occupations, e.g. in administration, arts, music, military etc. If we find that the effect is exclusive or at least far more pronounced for scientists, rather than for say generals, officers, artists or composers, this provides strong evidence that it was indeed the education aspect of the reforms, and not others driving our result.

Finally, it may be important to include additional control variables in the regression analysis to ensure even stronger balance between the treated and comparison states. We plan to include a range of other socio-economic characteristics for both Prussia and other German states during the 19th century. For Prussia, we use data from the Ifo Prussian Economic History Database (iPEHD), a resource offering an extensive collection of variables for all Prussian counties during this period (Hornung 2014). For the other states, we plan to collect the additional data from relevant sources. To account for educational infrastructure prior to the reforms, we will also include data on universities within each state or county, using data provided by De la Croix and Licandro (2015) and De la Croix et al. (2024).

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Appendix A. Regressions appendix

This appendix section presents additional regression results on different samples constructed based on the prominent people dataset from Schich et al. (2014).

TABLE A1. Effect of Prussian education reforms on academic births (1730–1855). Prominent academics born in Germany

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth) *Sample: All prominent academics born in German states*			
Panel A – Baseline			
Prussia × Post-1790	-0.051	0.031	0.031
	(0.036)	(0.041)	(0.041)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.359	0.393	0.400
Wild bootstrap p-value (DiD)	0.299	0.537	0.537
Panel B – Cities with pre-1790 academic births			
Prussia × Post-1790	0.036	-0.034	-0.034
	(0.057)	(0.065)	(0.066)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	3,822	3,796	3,796
R^2	0.353	0.401	0.413
Wild bootstrap p-value (DiD)	0.579	0.671	0.671
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	0.052	0.046	0.046
	(0.068)	(0.065)	(0.066)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.402	0.464	0.481
Wild bootstrap p-value (DiD)	0.501	0.519	0.519

Note. The dependent variable is the number of prominent academics born in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

TABLE A2. Effect of Prussian education reforms on academic births (1730–1855). All prominent academics born and died in Germany

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth) *Sample: All prominent academics born and died in German states*			
Panel A – Baseline			
Prussia × Post-1790	-0.030	0.037	0.037
	(0.028)	(0.033)	(0.033)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.328	0.359	0.367
Wild bootstrap p-value (DiD)	0.390	0.401	0.401
Panel B – Cities with pre-1790 academic births			
Prussia × Post-1790	0.044	-0.008	-0.008
	(0.048)	(0.056)	(0.056)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	3,406	3,380	3,380
R^2	0.329	0.373	0.390
Wild bootstrap p-value (DiD)	0.464	0.896	0.896
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	0.071	0.066	0.066
	(0.053)	(0.052)	(0.052)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.374	0.430	0.448
Wild bootstrap p-value (DiD)	0.317	0.325	0.325

Note. The dependent variable is the number of prominent academics born in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

A.1. Sample 1 (All): Prominent academics born in German states

A.2. Sample 2 (Stayers): Prominent academics born and died in German states

A.3. Sample 3 (Patriots): Prominent academics born and died in the same state

A.4. Sample 4 (Migrants): Prominent academics born and died in different German states

TABLE A3. Effect of Prussian education reforms on academic births (1730–1855). All prominent academics born and died in the same German state

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth) *Sample: All prominent academics born and died in the same German state*			
Panel A – Baseline			
Prussia × Post-1790	0.036***	0.071***	0.071***
	(0.012)	(0.015)	(0.015)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.290	0.309	0.317
Wild bootstrap p-value (DiD)	0.083	0.083	0.083
Panel B – Cities with pre-1790 academic births			
Prussia × Post-1790	0.120***	0.094***	0.094***
	(0.025)	(0.025)	(0.026)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,002	2,002	2,002
R^2	0.328	0.369	0.390
Wild bootstrap p-value (DiD)	0.048	0.023	0.023
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	0.113***	0.112***	0.112***
	(0.022)	(0.025)	(0.026)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.352	0.387	0.405
Wild bootstrap p-value (DiD)	0.034	0.012	0.012

Note. The dependent variable is the number of prominent academics born in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

TABLE A4. Effect of Prussian education reforms on academic births (1730–1855). Prominent academics born and died in different German states

	Column 1	Column 2	Column 3
Dependent variable: Number of births of prominent academics (AcadEducHealth) *Sample: All prominent academics born and died in different German states*			
Panel A – Baseline			
Prussia × Post-1790	-0.066**	-0.033	-0.033
	(0.031)	(0.031)	(0.031)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.214	0.232	0.248
Wild bootstrap p-value (DiD)	0.232	0.395	0.395
Panel B – Cities with pre-1790 academic births			
Prussia × Post-1790	-0.033	-0.072	-0.072
	(0.052)	(0.049)	(0.050)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,496	2,470	2,470
R^2	0.192	0.227	0.255
Wild bootstrap p-value (DiD)	0.591	0.394	0.394
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	-0.042	-0.046	-0.046
	(0.052)	(0.049)	(0.049)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.236	0.270	0.308
Wild bootstrap p-value (DiD)	0.501	0.437	0.437

Note. The dependent variable is the number of prominent academics born in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

Appendix B. Deaths of prominent academics by birth cohorts

B.1. Sample 1 (Stayers): Prominent academics born and died in Germany

B.2. Sample 2 (Patriots): Prominent academics born and died in the same state

B.3. Sample 3 (Migrants): Prominent academics born and died in different German states

TABLE A5. Effect of Prussian education reforms on academic deaths by birth cohorts (1730–1855). Prominent academics born and died in Germany.

	Column 1	Column 2	Column 3
Dependent variable: Number of deaths of prominent academics (AcadEducHealth) *Sample: All prominent academics born and died in Germany.*			
Panel A – Baseline			
Prussia × Post-1790	-0.054*	0.051	0.051
	(0.031)	(0.043)	(0.044)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.610	0.635	0.641
Wild bootstrap p-value (DiD)	0.212	0.443	0.443
Panel B – Cities with pre-1790 academic deaths			
Prussia × Post-1790	0.344***	0.342**	0.342**
	(0.109)	(0.139)	(0.141)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	1,742	1,716	1,716
R^2	0.606	0.653	0.665
Wild bootstrap p-value (DiD)	0.170	0.230	0.230
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	0.121	0.085	0.085
	(0.077)	(0.093)	(0.094)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.626	0.667	0.679
Wild bootstrap p-value (DiD)	0.274	0.431	0.431

Note. The dependent variable is the number of prominent academics died in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

TABLE A6. Effect of Prussian education reforms on academic deaths (1730–1855). All prominent academics born and died in the same German state

	Column 1	Column 2	Column 3
Dependent variable: Number of deaths of prominent academics (AcadEducHealth) *Sample: All prominent academics born and died in the same German state*			
Panel A – Baseline			
Prussia × Post-1790	0.035***	0.100***	0.100***
	(0.011)	(0.025)	(0.025)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.582	0.606	0.610
Wild bootstrap p-value (DiD)	0.085	0.205	0.205
Panel B – Cities with pre-1790 academic births			
Prussia × Post-1790	0.556***	0.732***	0.732***
	(0.059)	(0.116)	(0.119)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	1,248	1,248	1,248
R^2	0.594	0.648	0.658
Wild bootstrap p-value (DiD)	0.007	0.073	0.073
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	0.205***	0.189***	0.189***
	(0.024)	(0.046)	(0.046)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.602	0.641	0.649
Wild bootstrap p-value (DiD)	0.007	0.016	0.016

Note. The dependent variable is the number of prominent academics died in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

TABLE A7. Effect of Prussian education reforms on academic deaths (1730–1855). Prominent academics born and died in different German states

	Column 1	Column 2	Column 3
Dependent variable: Number of deaths of prominent academics (AcadEducHealth) *Sample: All prominent academics born and died in different German states*			
Panel A – Baseline			
Prussia × Post-1790	-0.089***	-0.050*	-0.050*
	(0.025)	(0.029)	(0.029)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	7,722	7,592	7,592
R^2	0.469	0.485	0.497
Wild bootstrap p-value (DiD)	0.053	0.225	0.225
Panel B – Cities with pre-1790 academic births			
Prussia × Post-1790	0.076	-0.135	-0.135
	(0.093)	(0.109)	(0.112)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	1,196	1,170	1,170
R^2	0.452	0.485	0.519
Wild bootstrap p-value (DiD)	0.504	0.512	0.512
Panel C – Cities above 5,000 residents			
Prussia × Post-1790	-0.084	-0.104	-0.104
	(0.062)	(0.063)	(0.064)
City & Cohort FE	✓	✓	✓
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	Yes
# Obs	2,626	2,626	2,626
R^2	0.492	0.518	0.545
Wild bootstrap p-value (DiD)	0.307	0.243	0.243

Note. The dependent variable is the number of prominent academics died in a city within a five-year birth cohort. All regressions include city and cohort fixed effects and cover cohorts starting in 1730–1855. The reported coefficient is on Prussia × Post-1790, where Post-1790 equals one for cohorts starting in 1790 or later. Column 1 includes no population controls. Columns 2 and 3 interact log population in 1750 and log population growth between 1700 and 1750 with the post indicator and cohort indicators, respectively. Panel A uses the full sample; Panel B restricts the sample to cities with at least one recorded academic birth before 1790; Panel C restricts the sample to cities with baseline population above 5,000 residents. Standard errors clustered by state are reported in parentheses. Stars indicate conventional clustered significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Wild cluster bootstrap p-values for the reported coefficient use 9,999 replications clustered by state. Bootstrap inference should be interpreted cautiously because Prussia is the only treated state.

Appendix C. State-level descriptive analysis

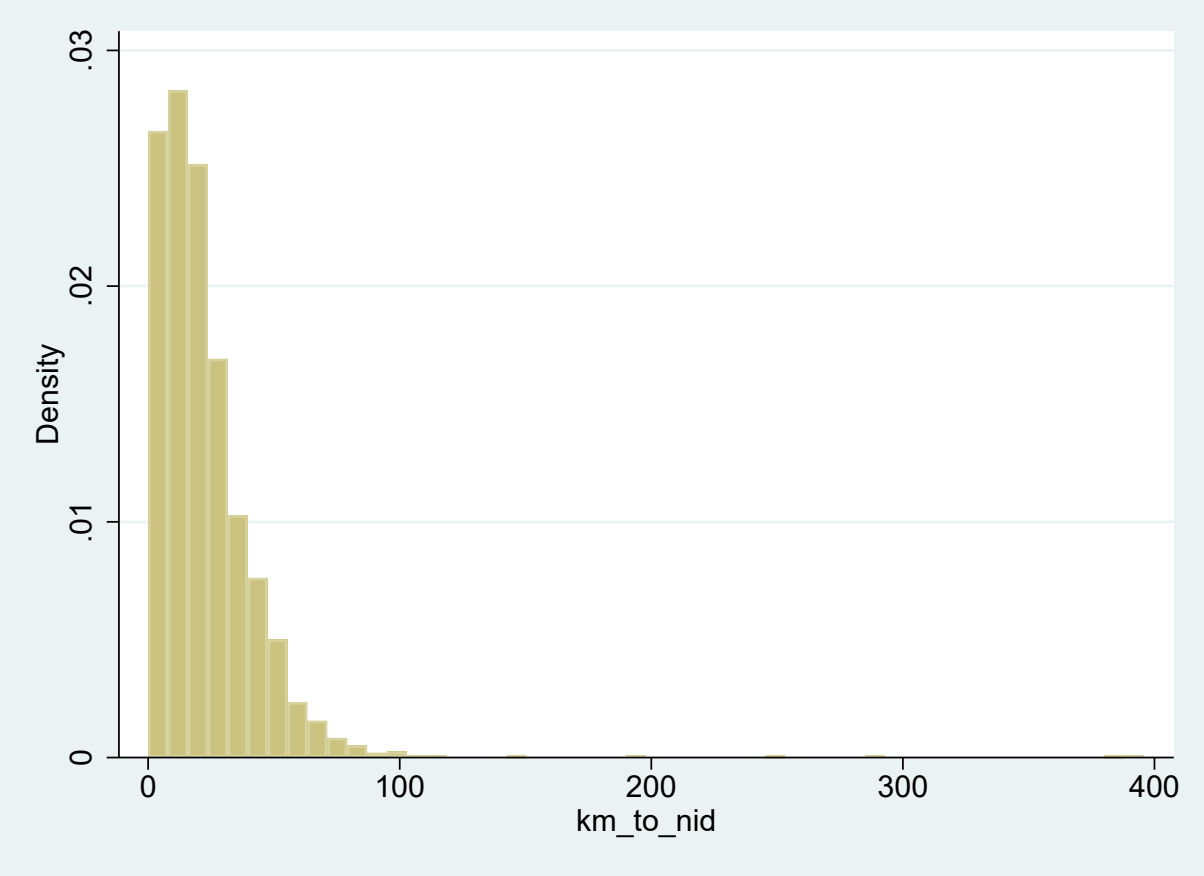


FIGURE A1. Calculated distance from birth location to nearest city, in km.

Note. This figure shows the distribution of the distance from a prominent person's birth location to the matched nearest neighboring city (restricted to be within the same state), using 50 bins. Most of the birth locations are matched to the cities withing 100 km distance. The 95th percentile is 55.71 km (2,298 obs out of 2,419 are below it); the 99th percentile is 84.27 km (2,395 obs out of 2,419 are below it).

Appendix D. Figures appendix

Appendix B presents some additional figures.

D.1. Larger figure, without panel, in the appendix

Here is a large, simple figure in the appendix (see figure **??**).

Appendix E. Robustness checks

E.1. Panel dataset from (De la Croix and Licandro 2015)

E.2. Subsection with references

Here is a sentence with some references in text: **??** found something but that is an uncommon result. The references can also go in parentheses at the end of the sentence (**??**). The references

go to the reference list at the end of the main text—so appendix and main text share a common list of references.

E.3. Subsection with a foonote

Here is a sentence with a footnote.[9] The numbering of the footnotes, just like the numbering of the pages, continues from the main text to the appendix.

Appendix F. Last appendix with text

This is a last appendix with a bit of text.

F.1. A first subsection

F.2. A second subsection

F.3. A third subsection

[9] Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt.